

Contributions: five named things that travel back into the field

Three theoretical · one methodological · one empirical-discovery

THEORETICAL (×3)

Propagation Operating Characteristic

A measured curve relating driven contraction (λ_1) to memory capacity. Identifies the operating region in which a driven reservoir is jointly stable and information-preserving.

Established: Chapter 6. Used to derive $\beta = 0.05$ in Exp C.

Measurement-Instrument Paradigm

A methodological commitment to treat the reservoir as a measurement instrument whose properties (ρ , λ_1 , MC, Taken-sufficiency) are characterized before use — not as a black-box learner.

Established: Chapters 3 and 6. Operationalized in Exp A / B / C.

Spike-to-Embedding Pipeline

A reproducible, interpretable transformation from raw ERP to a low-dimensional embedding via BSC₆ and PCA-64, with measured sufficiency for downstream classification.

Established: Chapter 4.

METHODOLOGICAL (×1)

Layer Ablation Methodology

A per-disorder, per-layer ablation protocol that isolates the contribution of each architectural level to clinical separability, tested against a channel-permutation null.

Established: Chapter 5. Tested in Exp D.

EMPIRICAL DISCOVERY (×1)

Centered Baseline Comparison

A clinical-cohort comparison against a centered, demographically-matched baseline (not an arbitrary control), producing the SUD and PTSD findings that survive permutation testing.

Established: Chapter 5. Tested in Exp D.

"These five names — the POC, the MIP, the spike-to-embedding pipeline, the layer ablation methodology, the centered baseline comparison — are this dissertation's contributions. They travel back into the field with my name on them."